Oxford Prospects and Global Development Institute

Course: STEM: Maths, Physics, Computer Science and Engineering

Session Title: Academic Profile Development

Speaker: Professor Martin Bureau and Eric Liang

Academic Assistant: Florence Goodrich

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Learning Objectives:
Developing your academic profile, and understanding more about British Postgraduate studies
Outline:
MSc Programmes PhD Programmes Student/Supervisor relationship Admission Process Career Paths Keys to Success
Main Points:
Masters In China, the classic master's degree is two years. This form of master's is tending to disappear because people are questioning the usefulness of the master's where you are going to need a PhD later in order to do academic work. There are now 1 year taught programmes, or integrated/fast-track master's programmes which move straight into a PhD. **PhD** It is good to know what a PhD really is. There can be theoretical PhDs or experimental PhDs. These days, a theoretical PhD tends to be running large simulations. You can also have instrumentation PhDs, which involves building apparatus, for example building telescopes or building software. You will need to read a lot of research papers and piece these together to get a good overview of the research field. You will need to discuss ideas, results, and progress with your supervisors, as well as other PhD students and collaborators. You will need to present your results, for example via papers and talks. There is no point doing good research if nobody knows about it. A PhD requires a wide range of skills, which can be quite different to the skills needed at undergrad. There may be taught courses, usually there are very few in the UK/Australia, but there are more in the US. Towards the end of your PhD, you will need to write and defend your thesis. In the sciences, this is often a lot of papers.

Throughout your PhD, you can get teaching, outreach, and supervision experience, and do networking. You also need to live your life, try new things, and experience different cultures.

Differences across countries

There are large differences between different countries, so research well before deciding where to study and applying. The admission times are at different times of year – for example Australian programmes start in March, UK programmes in October, and US programmes in September.

In the UK, you can do a PhD in 3 years. In the USA, the PhD is 6-7 years. You may therefore want to choose carefully where to study, as this will affect how long the PhD takes you. The nature of the work will be different between countries – in the US, you will have 2 years of coursework before starting, in the UK, this is not the case.

You need to select your supervisor carefully, as you will have to work with them closely. In the US, during your coursework you can 'test drive' professors by chatting and doing mini research projects with them. You can then decide on a supervisor later. In the UK, you decide your supervisor and project before you are admitted.

You can also do a PhD in a national research facility. This could be a very different environment, with only your topic studied and with no undergraduate students present. You need to consider where you want to live, for example in an urban or rural environment.

You most likely need to fund your degree. You could be funded by the government, by the university you are going to, or by the supervisor. There are costs with tuition, accommodation, food, etc. This is a big factor to think about.

The immigration requirements are also something to think about. For example, if you are studying something to do with national security, you may need special permission to work on that topic.

There are many factors to think about. You need to think about the language, the culture, the climate. Are there things you enjoy, for example scuba diving, that you can only do in certain countries? Choose for you, and not for your parents. It may be a choice that affects the rest of your life.

What is expected of you?

Dedication. There will probably be long hours, especially if you go to a prestigious university. You need good time management so that even though you work 40/50 hours a week you can still have a good work/life balance.

You will need good proficiency with English, and with technical writing. Most scientific communication occurs through papers, and these papers are written in English, so you really need to invest time in this. Learning how to code is also useful in the sciences.

You will need flexibility, the ability to learn new skills, and well as to be able to work as part of a team.

No two PhDs are the same, the projects, the teams and locations are different. It is always true that it is never going to be like your undergraduate programme. You will need to identify the question to research, and then piece together what you know, and what you need to find out.

What can be expected from your supervisor?

Similarly to students, they have long hours, need to time manage and be proficient in English. Professors have other obligations, so often don't have time to help. But they will give you advice when they can on what you should be doing.

Relationship between Student and Supervisor

You need to adapt yourself to the life and working conditions. The relationship between the student and supervisor is very different to the lecturer-student relationship during an undergraduate degree. In undergrad, the lecturer only speaks about the science in the textbooks. You don't speak about new or novel science together.

Now, you no longer work on coursework, you are working as scientific collaborator. You may also work on future projects after you graduate, and you will need reference letters after you graduate. Is helpful if you can collaborate well, and you can work efficiently.

It is your own project, so seek advice on it, but you can also get advice from your supervisors on other things, for example networking and your future career. Your supervisor can give you advice on where to apply to after you graduate from your PhD, and what you need to bear in mind for that transition.

You need to know what your supervisor is strong at, for example if coding is not a strength, the supervisor may not be able to help. You may consult a post doc or another student. Know who to ask the most efficient questions.

Often you may have an idea, and research it and find that the idea does not work. In the same way, you don't need to take everything your supervisor says as true, because they are just ideas.

Be brave if you need external help – your department and university will offer help if you need. Be resilient to harsh criticism of the science, and it is different from bullying.

A PhD students typical day -
- Browse the latest papers. Pick a few papers a week to read very carefully, to understand what is going on and use in your own research.
- Attend a few seminars/lectures to broaden knowledge of the field.
- Have a few meetings with collaborators

- Prepare for reports/papers/presentations
- Code (data analysis) – you may have to run simulations and analyse data for your research. You may also be doing experiments if you are on an experimental PhD.

You need to be very flexible – you need to balance many things, have excellent time management, and a priority ranking of things to do. You also need to balance having fun and enjoying life, and your health, as well. Find the most effective way to organise these.
You need to devote yourself to this project/career in order to succeed. The PhD career is demanding.

Have patience and perseverance, especially at the beginning. Recall your motivation for doing research in science. You need to know that the boring coding period is for the beautiful science moment in the end.

You need both individual and deep investigation into a question, as well as talking with people, and having active and synoptic discussion on broad topics with colleagues.

Studying abroad is hard work. You may not automatically be recognised when you get home, as a lot of people go abroad for their PhDs. It has many benefits in other ways, it can be mind-opening and indispensable. Whenever you meet with a hardship, talk to your peers and your supervisor. Talking to peers shows that everyone is experiencing the same, you are not alone. Your supervisor can give you really helpful advice.

Application process

You need to find out what the timescale is, as sometimes there can be earlier deadlines if you want to apply for a scholarship. You may need to submit your undergraduate grades and transcripts. You may need proof of language proficiency. In the US, you may need a GRE. You will need a CV. In China there is a large focus on getting good grades in exams, and this focus is not so strong in the west. They will also want to see other things on your CV outside of the exams. This will show you have more generic skills such as time management, leadership etc. Show if you have been a captain of a sports team, if you have any research experience, if you have other skills.
In terms of references, they will want to know what can't be seen in the other elements of the application, for example how you are as a person, if you are hard working or organised for example.

Contact potential supervisors, but be sure to personalise the email to them, show that you are actually interested in them, and don't just send a generic email. A supervisor is more likely to reply to a personalised email. For example, say I looked at your website, and I read your papers and I'm interested.
If you make it to interviews, you need to practice so that you get good at interviews. This practice can be with friends or teachers.

An application from a student with funding is much more attractive.

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Engagement is much more important than accomplishment – everybody has good grades. They need other ways to tell people apart. A stand-alone MSc with research (may have a thesis + paper) can show that you are engaged, and that you have research experience.

You need a breadth of achievement – not only academic.
You need interpersonal skills because you will be working with other people.
There is no discrimination for gender, nationality, etc.
Be sure to apply to institutes covering a range of difficulties – apply to some you are sure you will get in, and some that are aspirational.

Student background, Eric Liang –

He did a master's with a supervisor that he already knew from his undergraduate degree in order to gain research experience. He chose the institutes to apply to for a PhD mainly based on the fields, and where had a good research group. He applied to about 10, which is typical, and consulted the ranking in order to decide where to apply.

He chose the supervisor based on the field, and the project the supervisor is offering.
All departments will post potential research projects on website, and then you can contact the supervisor beforehand.

He did not get an agency to organise his applications. Practice interviews, practice with yourself, and with professors from your undergraduate university.

Be careful with your eligibility, and check you are eligible to apply. This can be vague on websites sometimes but contact staff to check.

Common merits during application –

Be honest and clear, do not attempt to cheat during your interview. Your integrity is very important. You do not need every item of your application to be perfect, but you need some, in order to make you outstanding and impressive. Be concise, the interviewing committee will read a lot of material. Be academic, academic writing is something you need to work on, as it is a different skill. Be confident, and relaxed. Be resilient. Take every application seriously, but do not take them too seriously. Appreciate each institution you apply to. You do not need to be perfect at your first interview but improve them as you go. Let people know what you are strong at, let people know what you can offer, but be honest and resilient.

Career Paths

You could go into a job, or you could do a PhD straight away. If you are not sure, you could do a master's, which might help you to decide whether you want to do a PhD or not.

There are many factors to consider – there are personal and financial conditions. There are also personal vs. professional interests to think about, for example you may have a partner that doesn't want to go abroad. There are also the actual opportunities to consider.

After your PhD –
It is common to do 2 or 3 post-doctoral positions, which are 2-5 years each. Once you have landed a job, you tend to be on a tenure track, for 5 years. Then if you are successful in that, you land a permanent job. It is a long path. But post docs are normal jobs, and you can do them in nice places, so it can be good.

As a post doc there can be a wide variety of jobs, for example research, supervising students, collaborating with others, teaching, public outreach, applying for funding to get grants, governance, for example of the department, facilities etc, citizenship, helping on industry wide projects, for example launching a new telescope.

Keys to Success

Control what you can – know what you desire and know what your goals are.
Ensure that you do what you enjoy. Martin went to universities that were good, but not necessarily the best, but they allowed him to live in places that he enjoyed and he wanted to live. Then if the PhD failed, he was still happy. The point is, make decisions which are good professionally, but also good personally. You also need to be lucky, he happened to end up in Oxford because there happened to be a job at the right time. You need to create your luck though, be in the right places at the right time, build the right instrument etc. Take calculated risks, for example where Martin went to the Netherlands, there was an emerging technology he worked on, which became popular, and then suddenly everyone in his team was very desirable.
Enjoy it whilst it lasts but have good back up plans.

A big skill is to be able to identify the right questions, and the right tools to work with them. Surround yourself with excellent people and facilities. You need to have access to the right resources. Work hard. Create your luck. And follow where the field is going.

Know yourself – know what you want to do, know your desires. Balance your desires with the reality in the world, find what the opportunities are out there. The opportunities will vary year on year. You need to concentrate on what your most important thing is. And be able to prioritise and make it successful.

Discussion

Huang Jianjun – I major in Journalism and minor in Computer Science, am I qualified to apply for a Master's in Engineering?

Martin – he suspects the answer is no but look at the selection criteria for a specific university. In the UK, the answer is no, because we don't have majors and minors. Best bet is to look at US and then look university to university.

Zhou Danni – Will a part time internship in a big-name company help when applying for MSc?

Martin - Yes, it will definitely help, anything they you get that differentiates you from others will help.

Xie Lulin – Are there rigid requirements about writing papers during your PhD degree?

Martin – There is no formal requirement, but the expectation is probably about 3 first author papers during your PhD. If you are doing instrumentation, this will not happen, will be more reports on the instruments. Generally, 3 is common, one per year. The requirement is a thesis, so if you have written 3 core chapters with results but they are not published, this may be ok as there is no official requirement. The only formal requirement is that you have generated original, impactful, high-quality research.

Yidi Wang/Zhou Danni – In order to get a good grade, they feel they should spend most of the semester preparing for exams. They should also spend time on research-based things. How do you balance time between preparing for exams and doing research-based things if he wants to apply for a PhD?

Martin – Whether you come number 1 or 5 or 10 in exams is not important, clearly you need to do reasonably well in your exams. But extracurricular work is more important. Often people do this doing summer internships, which you can do in China, or abroad, for example at CERN. Work hard, find programmes, and take the initiative.

Eric – You can use the summer and winter vacation to do research, because during term time you have time to do exams, as well as to have fun.

Xu Jie – Is it helpful to participate in an exchange programme in your last undergraduate year?

Eric – This could be a good period of time to carry out research. You don't need to publish but it is a good time to get experience.
Martin – Bear in mind that it will be too late if you have already applied to graduate school.

He Yixin – How important is the recommendation letter for the application?

Martin – It is very important. The letter is the only bit of information where they get to see about you. What they want to see is how you are as a person, for example, are you hard working.
It is important to have an objective research letter. For example, one where the person has nothing to gain from saying you are really good.

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Sun Zhe – Is the minor degree useless when applying to the UK?
Martin – Yes because a minor degree does not exist in the UK. If you have a major and a minor in science, then it can be helpful, but look at the sum of the major and the minor, and this is what the UK will look at.

How difficult was it to make friends, did you experience culture shock?
Eric - He didn't experience too much culture shock because he had been abroad already for a summer project, so he was a bit used to it. You also have access to foreign materials which helps, for example TV dramas. Many Chinese students only make friends with Chinese students, but he doesn't recommend this, as you will experience more if you make friends with foreign people. There are also more students from overseas than from the UK, so these people are also sharing similar experiences to you, so you have a lot to talk about.

Martin – Everybody is in the same boat, they are all starting fresh, and everyone is looking to make friends. It is not too difficult to find friends.

Zhu Yantong – Many internships prefer people who have research experience, so what should we do to be competitive the first time?

Martin – It is difficult but try to sell yourself. There are other ways to acquire experience, for example teaching music to younger students, or running a club. You have to be creative.